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Inventor(s)	SHIMOYAMA; Masashi et al.

PLATING APPARATUS AND PLATING METHOD

Abstract

A specific portion of a substrate is allowed to be shielded at a desired timing, and making a plating film-thickness uniform is improved.
A plating module includes: a plating tank **410** for housing a plating solution; an anode **430** disposed in the plating tank **410**; a substrate holder **440** for holding a substrate Wf with a surface to be plated Wf-a facing downward; a rotation mechanism **447** configured to rotate the substrate holder **440** in a first direction and a second direction opposite to the first direction; and a shielding mechanism **485** configured to move a shielding member **481** into between the anode **430** and the substrate Wf depending on a rotation angle of the substrate holder **440.**

Inventors: **SHIMOYAMA; Masashi (Tokyo, JP), MASUDA; Yasuyuki (Tokyo, JP), HIWATASHI; Ryosuke (Tokyo, JP)**

Applicant: **EBARA CORPORATION** (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED MATTERS [0001] This application is a divisional of U.S. patent application Ser. No. 17/923,163, filed Nov. 3, 2022, which is a U.S. national phase application of International Patent Application No. PCT/JP2022/003526 filed Jan. 31, 2022, the entire contents of all foregoing applications are hereby incorporated by reference for any and all purposes.

TECHNICAL FIELD
[0002] This application relates to a plating apparatus and a plating method.
BACKGROUND ART
[0003] As an example of a plating apparatus, there has been known a cup type electroplating apparatus. The cup type electroplating apparatus immerses a substrate (for example, semiconductor wafer) held by a substrate holder with a surface to be plated facing downward in a plating solution, and applies a voltage between the substrate and an anode, thereby depositing a conductive film on a surface of the substrate.
[0004] It has been known that the cup type electroplating apparatus uses a shielding member to seal an electric field formed between the anode and the substrate. For example, PTL 1 discloses an electroplating apparatus configured to shield a specific portion of a substrate only at a desired timing by moving a shielding member into between the specific portion of the substrate and an anode when the specific portion of the substrate is rotated within a predetermined range of a rotation angle.
CITATION LIST
Patent Literature
[0005] PTL 1: Japanese Patent No. 6901646
SUMMARY OF INVENTION
Technical Problem
[0006] However, the electroplating apparatus of the prior art is desired to uniform a plating film-thickness by enhancing a stirring strength of a plating solution housed in a plating tank while shielding a specific portion of a substrate at a desired timing.
[0007] That is, in the prior art, since a substrate holder is rotated in one direction at a constant rotation speed, a time period of shielding the specific portion of the substrate is fixed. In this respect, for example, when it is desired to shield the specific portion of the substrate for a longer time, it is considered to decrease the rotation speed of the substrate holder while the specific portion of the substrate rotates in a predetermined range of a rotation angle. However, decreasing the rotation speed of the substrate holder reduces the stirring strength of the plating solution housed in the plating tank, and consequently, making the thickness of the plating film formed on a surface to be plated uniform is possibly hindered.
[0008] Therefore, this application has an object to achieve a plating apparatus and a plating method allowing to shield a specific portion of a substrate at a desired timing and allowing to improve making a plating film-thickness uniform.
Solution to Problem
[0009] According to one embodiment, a plating apparatus is disclosed. The plating apparatus includes: a plating tank for housing a plating solution; an anode disposed in the plating tank; a substrate holder for holding a substrate with a surface to be plated facing downward; a rotation mechanism configured to rotate the substrate holder in a first direction and a second direction

opposite to the first direction; and a shielding mechanism configured to move a shielding member into between the anode and the substrate depending on a rotation angle of the substrate holder.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a perspective view illustrating an overall configuration of a plating apparatus of the embodiment.

[0011] FIG. **2** is a plan view illustrating the overall configuration of the plating apparatus of the embodiment.

[0012] FIG. **3** is a vertical cross-sectional view schematically illustrating a configuration of a plating module of one embodiment, and illustrates a state where a shielding member is moved to a retracted position.

[0013] FIG. **4** is a vertical cross-sectional view schematically illustrating the configuration of the plating module of one embodiment, and illustrates a state where the shielding member is moved to a shielding position.

[0014] FIG. **5** is a perspective view schematically illustrating a configuration of a shielding mechanism of one embodiment.

[0015] FIG. **6** is a perspective view schematically illustrating the configuration of the shielding mechanism of one embodiment.

[0016] FIG. **7** includes plan views schematically illustrating the configuration of the shielding mechanism of one embodiment.

[0017] FIG. **8** is a drawing illustrating a relation between a timing of shielding a specific portion of a substrate and a rotation speed of a substrate holder.

[0018] FIG. **9** is a drawing illustrating a relation between the timing of shielding the specific portion of the substrate and the rotation speed of the substrate holder.

[0019] FIG. **10** is a drawing illustrating a relation between the timing of shielding the specific portion of the substrate and the rotation speed of the substrate holder.

[0020] FIG. **11** is a perspective view schematically illustrating a configuration of a shielding mechanism of one embodiment.

[0021] FIG. **12** is a perspective view schematically illustrating the configuration of the shielding mechanism of one embodiment.

[0022] FIG. **13** is a perspective view schematically illustrating a part of the configuration of the shielding mechanism of one embodiment.

[0023] FIG. **14** includes plan views schematically illustrating the configuration of the shielding mechanism of one embodiment.

[0024] FIG. **15** is a perspective view schematically illustrating the configuration of the shielding mechanism of one embodiment.

[0025] FIG. **16** includes plan views schematically illustrating the configuration of the shielding mechanism of one embodiment.

[0026] FIG. **17** is a vertical cross-sectional view schematically illustrating a configuration of a plating module of one embodiment, and illustrates a state where a shielding member is retracted.

[0027] FIG. **18** is a top view schematically illustrating the configuration of the plating module of one embodiment, and illustrates the state where the shielding member is retracted.

[0028] FIG. **19** is a vertical cross-sectional view schematically illustrating the configuration of the plating module of one embodiment, and illustrates a state where the shielding member is moved into between an anode and a substrate.

[0029] FIG. **20** is a top view schematically illustrating the configuration of the plating module of one embodiment, and illustrates the state where the shielding member is moved into between the

anode and the substrate.
[0030] FIG. **21** is a vertical cross-sectional view schematically illustrating a configuration of a plating module of one embodiment.
[0031] FIG. **22** is a flowchart of a plating method using the plating module of one embodiment.
[0032] FIG. **23** is a flowchart of a plating method using the plating module of one embodiment.
DESCRIPTION OF EMBODIMENTS
[0033] The following describes embodiments of the present invention with reference to the drawings. In the drawings described below, identical reference numerals are attached to identical or equivalent components, and overlapping description will be omitted.
<Overall Configuration of Plating Apparatus>
[0034] FIG. **1** is a perspective view illustrating the overall configuration of the plating apparatus of this embodiment. FIG. **2** is a plan view illustrating the overall configuration of the plating apparatus of this embodiment. As illustrated in FIGS. **1** and **2**, a plating apparatus **1000** includes load ports **100**, a transfer robot **110**, aligners **120**, pre-wet modules **200**, pre-soak modules **300**, plating modules **400**, cleaning modules **500**, spin rinse dryers **600**, a transfer device **700**, and a control module **800**.
[0035] The load port **100** is a module for loading a substrate housed in a cassette, such as a FOUP, (not illustrated) to the plating apparatus **1000** and unloading the substrate from the plating apparatus **1000** to the cassette. While the four load ports **100** are arranged in the horizontal direction in this embodiment, the number of load ports **100** and arrangement of the load ports **100** are arbitrary. The transfer robot **110** is a robot for transferring the substrate that is configured to grip or release the substrate between the load port **100**, the aligner **120**, the pre-wet module **200**, and the spin rinse dryers **600**. The transfer robot **110** and the transfer device **700** can perform delivery and receipt of the substrate via a temporary placement table (not illustrated) to grip or release the substrate between the transfer robot **110** and the transfer device **700**.
[0036] The aligner **120** is a module for adjusting a position of an orientation flat, a notch, and the like of the substrate in a predetermined direction. While the two aligners **120** are disposed to be arranged in the horizontal direction in this embodiment, the number of aligners **120** and arrangement of the aligners **120** are arbitrary. The pre-wet module **200** wets a surface to be plated of the substrate before a plating process with a process liquid, such as pure water or deaerated water, to replace air inside a pattern formed on the surface of the substrate with the process liquid. The pre-wet module **200** is configured to perform a pre-wet process to facilitate supplying the plating solution to the inside of the pattern by replacing the process liquid inside the pattern with a plating solution during plating. While the two pre-wet modules **200** are disposed to be arranged in the vertical direction in this embodiment, the number of pre-wet modules **200** and arrangement of the pre-wet modules **200** are arbitrary.
[0037] For example, the pre-soak module **300** is configured to remove an oxidized film having a large electrical resistance present on, a surface of a seed layer formed on the surface to be plated of the substrate before the plating process by etching with a process liquid, such as sulfuric acid and hydrochloric acid, and perform a pre-soak process that cleans or activates a surface of a plating base layer. While the two pre-soak modules **300** are disposed to be arranged in the vertical direction in this embodiment, the number of pre-soak modules **300** and arrangement of the pre-soak modules **300** are arbitrary. The plating module **400** performs the plating process on the substrate. There are two sets of the 12 plating modules **400** arranged by three in the vertical direction and by four in the horizontal direction, and the total 24 plating modules **400** are disposed in this embodiment, but the number of plating modules **400** and arrangement of the plating modules **400** are arbitrary.
[0038] The cleaning module **500** is configured to perform a cleaning process on the substrate to remove the plating solution or the like left on the substrate after the plating process. While the two cleaning modules **500** are disposed to be arranged in the vertical direction in this embodiment, the

number of cleaning modules **500** and arrangement of the cleaning modules **500** are arbitrary. The spin rinse dryer **600** is a module for rotating the substrate after the cleaning process at high speed and drying the substrate. While the two spin rinse dryers are disposed to be arranged in the vertical direction in this embodiment, the number of spin rinse dryers and arrangement of the spin rinse dryers are arbitrary. The transfer device **700** is a device for transferring the substrate between the plurality of modules inside the plating apparatus **1000**. The control module **800** is configured to control the plurality of modules in the plating apparatus **1000** and can be configured of, for example, a general computer including input/output interfaces with an operator or a dedicated computer.

[0039] An example of a sequence of the plating processes by the plating apparatus **1000** will be described. First, the substrate housed in the cassette is loaded on the load port **100**. Subsequently, the transfer robot **110** grips the substrate from the cassette at the load port **100** and transfers the substrate to the aligners **120**. The aligner **120** adjusts the position of the orientation flat, the notch, or the like of the substrate in the predetermined direction. The transfer robot **110** grips or releases the substrate whose direction is adjusted with the aligners **120** to the pre-wet module **200**.

[0040] The pre-wet module **200** performs the pre-wet process on the substrate. The transfer device **700** transfers the substrate on which the pre-wet process has been performed to the pre-soak module **300**. The pre-soak module **300** performs the pre-soak process on the substrate. The transfer device **700** transfers the substrate on which the pre-soak process has been performed to the plating module **400**. The plating module **400** performs the plating process on the substrate.

[0041] The transfer device **700** transfers the substrate on which the plating process has been performed to the cleaning module **500**. The cleaning module **500** performs the cleaning process on the substrate. The transfer device **700** transfers the substrate on which the cleaning process has been performed to the spin rinse dryer **600**. The spin rinse dryer **600** performs the drying process on the substrate. The transfer robot **110** receives the substrate from the spin rinse dryer **600** and transfers the substrate, on which the drying process is performed, to the cassette at the load port **100**. Finally, the cassette housing the substrate is unloaded from the load port **100**.

<Configuration of Plating Module>

[0042] Next, the configuration of the plating module **400** will be described. Since the 24 plating modules **400** according to the embodiment have the same configuration, only one of the plating modules **400** will be described. FIG. **3** is a vertical cross-sectional view schematically illustrating a configuration of a plating module of one embodiment, and illustrates a state where a shielding member is moved to a position (referred to as a “retracted position” as necessary) apart from between an anode and a substrate. FIG. **4** is a vertical cross-sectional view schematically illustrating the configuration of the plating module of one embodiment, and illustrates a state where the shielding member is moved to a position (referred to as a “shielding position” as necessary) between the anode and the substrate.

[0043] As illustrated in FIG. **3** and FIG. **4**, the plating module **400** includes a plating tank **410** for housing a plating solution. The plating module **400** includes a membrane **420** dividing inside the plating tank **410** in the vertical direction. The inside of the plating tank **410** is divided into a cathode region **422** and an anode region **424** by the membrane **420**.

[0044] The cathode region **422** and the anode region **424** are each filled with the plating solution. The plating module **400** includes a nozzle **426** opening toward the cathode region **422**, and a supply source **428** for supplying the plating solution to the cathode region **422** via the nozzle **426**. For the anode region **424** as well, the plating module **400** includes a mechanism for supplying the plating solution to the anode region **424**, but it is not illustrated. An anode **430** is disposed on a bottom surface of the plating tank **410** in the anode region **424**. An ionically resistive element **450** is disposed to be opposed to the membrane **420** in the cathode region **422**. The ionically resistive element **450** is a member for attempting to uniformize a plating process on a surface to be plated Wf-a of a substrate Wf, and is configured of a plate-shaped member provided with many holes.

[0045] Further, the plating module **400** includes a substrate holder **440** for holding the substrate Wf with the surface to be plated Wf-a facing downward. The substrate holder **440** includes a power feeding contact point (not illustrated) for feeding power from a power source to the substrate Wf. The substrate holder **440** includes a seal ring holder **442** for supporting an outer edge portion of the surface to be plated Wf-a of the substrate Wf and a frame **446** for holding the seal ring holder **442** to a substrate holder main body (not illustrated). Further, the substrate holder **440** includes a back plate **444** for pressing a back surface of the surface to be plated Wf-a of the substrate Wf and a shaft **448** attached to a back surface of a substrate pressing surface of the back plate **444**.

[0046] The plating module **400** includes an elevating mechanism **443** for moving up and down the substrate holder **440** and a rotation mechanism **447** for rotating the substrate holder **440** so that the substrate Wf rotates about a virtual axis (virtual rotation axis extending perpendicularly in a center of the surface to be plated Wf-a) of the shaft **448**. The rotation mechanism **447** is configured to rotate the substrate holder **440** in a first direction (for example, clockwise) and a second direction (counterclockwise) opposite to the first direction. In other words, the rotation mechanism **447** can rotate the substrate holder **440** in the first direction, and can rotate the substrate holder **440** in the second direction by switching the rotation direction. The elevating mechanism **443** and the rotation mechanism **447** can be achieved by a known mechanism, such as a motor. The plating module **400** is configured to perform the plating process on the surface to be plated Wf-a of the substrate Wf by immersing the substrate Wf in the plating solution in the cathode region **422** using the elevating mechanism **443** and applying a voltage between the anode **430** and the substrate Wf.

[0047] The plating module **400** includes a shielding member **481** for shielding an electric field formed between the anode **430** and the substrate Wf when the shielding member **481** is arranged between the anode **430** and the substrate Wf. The shielding member **481** may be, for example, a shielding plate formed in a plate shape. The plating module **400** includes a shielding mechanism **485** for moving the shielding member **481**. The shielding mechanism **485** is configured to operate in response to a command signal based on information regarding a rotation angle of the substrate holder **440** input from the control module **800**. Specifically, the shielding mechanism **485** is configured to move the shielding member **481** to the retracted position as illustrated in FIG. **3** when the rotation angle of a specific portion of the substrate Wf is out of a predetermined range. Further, the shielding mechanism **485** is configured to move the shielding member **481** to the shielding position as illustrated in FIG. **4** when the rotation angle of the specific portion of the substrate Wf is within the predetermined range. That is, the shielding mechanism **485** is configured to linearly move the shielding member **481** between the retracted position and the shielding position depending on the rotation angle of the substrate holder **440**. The following describes a specific example of the shielding mechanism **485**.

[0048] FIG. **5** is a perspective view schematically illustrating a configuration of a shielding mechanism of one embodiment. FIG. **6** is a perspective view schematically illustrating the configuration of the shielding mechanism of one embodiment. FIG. **7** includes plan views schematically illustrating the configuration of the shielding mechanism of one embodiment. FIG. **7**A illustrates a state where the shielding member **481** is at the retracted position, and FIG. **7**B illustrates a state where the shielding member **481** is at the shielding position.

[0049] As illustrated in FIG. **5** to FIG. **7**, the shielding mechanism **485** includes a cam member **487**, a rotation drive mechanism **486** configured to rotate the cam member **487**, and a driven member **488** configured to linearly move the shielding member **481** between the shielding position and the retracted position in association with a rotation of the cam member **487**. The rotation drive mechanism **486** can be achieved by a known mechanism, such as a rotation motor.

[0050] The cam member **487** has a cam main body **487***b* configured to rotate by the rotation drive mechanism **486** and a rotor **487***a* attached to the cam main body **487***b*. The rotor **487***a* is attached to the cam main body **487***b* at a position eccentric with respect to a rotation axis of the rotation drive mechanism **486**.

[0051] The driven member **488** includes a driven slider **489** arranged on a pedestal **490-1** and a linear motion guide **490-2** configured to guide the driven slider **489**. On an upper surface of the pedestal **490-1**, a groove **490-1***a* is formed along a direction identical to a linear motion direction between the shielding position and the retracted position of the shielding member **481**. The driven slider **489** is arranged on the pedestal **490-1** via the linear motion guide **490-2** arranged in the groove **490-1***a*. The linear motion guide **490-2** is configured to guide the driven slider **489** along the groove **490-1***a*. This allows the driven slider **489** to reciprocate in the direction of the groove **490-1***a*. The driven slider **489** is arranged being opposed to the rotation drive mechanism **486** across the cam member **487**. On an opposed surface of the driven slider **489** to the rotation drive mechanism **486**, a cam groove **489***a* is formed along a vertical direction. The rotor **487***a* of the cam member **487** is fitted in the cam groove **489***a*. The shielding member **481** is attached to the driven slider **489** via a plate-shaped bracket **483** extending in the vertical direction.

[0052] When the rotation drive mechanism **486** rotates the cam member **487** (cam main body **487***b*), the rotor **487***a* rotates about the rotation axis of the rotation drive mechanism **486**. At this time, the rotor **487***a* presses a side surface of the cam groove **489***a*. This causes the driven slider **489** to move along the groove **490-1***a*. When the cam member **487** is rotated through a half turn (180 degree turn) from the state (retracted position) illustrated in FIG. **5** and FIG. **6**, the driven slider **489** moves the shielding member **481** to the shielding position. When the rotation drive mechanism **486** stops the rotation of the cam member **487** in this state, the shielding member **481** remains to be moved to the shielding position. Meanwhile, when the rotation drive mechanism **486** further rotates the cam member **487** through a half turn (180 degree turn) from this state, the driven slider **489** moves the shielding member **481** to the retracted position. That is, the driven slider **489** can linearly move the shielding member **481** between the shielding position and the retracted position by reciprocating along the groove **490-1***a* in association with the rotation of the cam member **487**.

[0053] The rotation drive mechanism **486** is configured to rotate the cam member **487** depending on the rotation angle of the substrate holder **440**. That is, the rotation drive mechanism **486** can rotate the cam member **487** so as to push out the shielding member **481** to the shielding position when the specific portion of the substrate Wf rotates within the predetermined angle range.

[0054] FIG. **8** is a drawing illustrating a relation between a timing of shielding a specific portion of a substrate and a rotation speed of a substrate holder. The horizontal axis of the graph of FIG. **8** indicates a rotation position of the specific portion of the substrate Wf, and the vertical axis indicates the position (shielding position or retracted position) of the shielding member and the rotation speed (rotation direction) of the substrate holder **440**. FIG. **8** illustrates the position of the shielding member and the rotation speed of the substrate holder when the substrate holder is rotated in the first direction (clockwise) at a constant speed. As illustrated in FIG. **8**, the substrate Wf is provided with a notch (cutout) Wf-n. In this example, as illustrated in FIG. **8**, assume that a specific portion α in which a deposition rate of plating is to be suppressed is present in a peripheral edge portion within a range of from $\theta=\theta 1$ to $\theta=\theta 2$ when the position of the notch Wf-n is a reference ($\theta=0$). Additionally, assume that the substrate Wf is arranged such that the notch Wf-n overlaps with the center of the shielding member **481** as illustrated in FIG. **8** in an initial state of starting the plating process, and starts rotating from the state. The specific portion α, θ**1**, and θ**2** can be determined based on the shape of the shielding member **481**, the shape of the specific portion α, and an intensity necessary for suppressing the deposition rate of plating. Therefore, in the relation between θ**1**, θ**2** and the specific portion α, θ**1**, θ**2** may be in the specific portion α, may be on a boundary of the specific portion α, and may be apart from the specific portion α.

[0055] The rotation mechanism **447** rotates the substrate holder **440** in the first direction at a predetermined speed as indicated by an arrow A in FIG. **8**. The shielding mechanism **485** (rotation drive mechanism **486**) pushes out the shielding member **481** to the shielding position when the position of θ**1** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the substrate holder **440** or the specific portion α is θ**1**). Subsequently, the shielding mechanism **485** (rotation drive mechanism **486**) pushes back the shielding member **481** to the retracted position when the position of θ**2** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**2**). Thus, the shielding mechanism **485** is configured to move the shielding member **481** between the anode **430** and the specific portion α of the substrate Wf when the rotation angle of the specific portion α of the substrate Wf is within a range of from θ**1** to θ**2**.

[0056] According to the embodiment, the specific portion α of the substrate Wf can be covered with the shielding member **481** at a desired timing. That is, according to the embodiment, since the specific portion α can be covered with the shielding member **481** at the desired timing instead of constantly covering the specific portion α with the shielding member **481**, an electric field in the specific portion α can be appropriately suppressed, and consequently, the deposition rate of plating in the specific portion α can be suppressed. While an example in which the specific portion α is covered with the shielding member **481** when the rotation angle of the specific portion α is within the predetermined range is described in this embodiment, it is not limited to this. For example, when the deposition rate of plating in the specific portion α is to be increased, the shielding member **481** can be retracted when the rotation angle of the specific portion α is within a predetermined range, and the shielding member **481** can be moved to the shielding position when the rotation angle of the specific portion α is out of the predetermined range.

[0057] As illustrated in FIG. **7** and the like, the shielding member **481** includes an arc-shaped mask member **481***a* corresponding to a part of the peripheral edge portion of the circular plate-shaped substrate Wf. Since the specific portion α is formed in an arc shape at the peripheral edge portion of the substrate Wf in some cases, only the specific portion α can be appropriately covered by covering the specific portion α of the substrate Wf using the arc-shaped mask member **481***a*. The same applies to embodiments below.

[0058] Additionally, in this embodiment, the rotation mechanism **447** is configured to switch the rotation direction of the substrate holder **440** between the first direction and the second direction when the rotation angle of the specific portion α of the substrate Wf is within a predetermined range (from θ**1** to θ**2**).

[0059] FIG. **9** is a drawing illustrating a relation between the timing of shielding the specific portion of the substrate and the rotation speed of the substrate holder. The horizontal axis of the graph of FIG. **9** indicates the rotation position of the specific portion of the substrate Wf, and the vertical axis indicates the position of the shielding member (shielding position or retracted position) and the rotation speed (rotation direction) of the substrate holder **440**. FIG. **9** illustrates the position of the shielding member and the rotation speed of the substrate holder when an additional suppression of the deposition rate of plating in the specific portion α of the substrate Wf is performed once (the rotation direction of the substrate holder is switched twice). As illustrated in FIG. **9**, the rotation mechanism **447** rotates the substrate holder **440** in the first direction at a predetermined speed at first as indicated by an arrow A in FIG. **9**. Subsequently, the rotation mechanism **447** pushes out the shielding member **481** to the shielding position when the position of θ**1** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**1**). Subsequently, the shielding mechanism **485** (rotation drive mechanism **486**) switches the rotation direction of the substrate holder **440** to rotate the substrate holder **440** in the second direction when the position of θ**2** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**2**). Subsequently, the rotation mechanism **447** switches the rotation direction of the substrate holder **440** to rotate the substrate holder **440** in the first direction when the position of θ**1** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**1**).

[0060] The shielding mechanism **485** (rotation drive mechanism **486**) switches the rotation direction of the substrate holder **440**, thereby allowing pushing out the shielding member **481** to the shielding position for a period of about three times as illustrated in FIG. **9** compared with a case where the substrate holder **440** is rotated in the first direction at the constant speed.

[0061] Accordingly, with the embodiment, the electric field in the specific portion α of the substrate Wf can be strongly suppressed. Additionally, according to the embodiment, since the stirring strength of the plating solution housed in the plating tank **410** can be enhanced by rotating the substrate holder **440** in the opposite direction, making the plating film-thickness uniform can be improved.

[0062] FIG. **10** is a drawing illustrating a relation between the timing of shielding the specific portion of the substrate and the rotation speed of the substrate holder. The horizontal axis of the graph of FIG. **10** indicates the rotation position of the specific portion of the substrate Wf, and the vertical axis indicates the position of the shielding member (shielding position or retracted position) and the rotation speed (rotation direction) of the substrate holder **440**. FIG. **10** illustrates the position of the shielding member and the rotation speed of the substrate holder when an additional suppression of the deposition rate of plating in the specific portion α of the substrate Wf is performed twice (the rotation direction of the substrate holder is switched four times). As illustrated in FIG. **10**, the rotation mechanism **447** rotates the substrate holder **440** in the first direction at a predetermined speed at first as indicated by an arrow A in FIG. **10**. Subsequently, the rotation mechanism **447** pushes out the shielding member **481** to the shielding position when the position of θ**1** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**1**). Subsequently, the shielding mechanism **485** (rotation drive mechanism **486**) switches the rotation direction of the substrate holder **440** to rotate the substrate holder **440** in the second direction when the position of θ**2** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**2**). Subsequently, the rotation mechanism **447** switches the rotation direction of the substrate holder **440** to rotate the substrate holder **440** in the first direction when the position of θ**1** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**1**).

[0063] Subsequently, the rotation mechanism **447** switches the rotation direction of the substrate holder **440** to rotate the substrate holder **440** in the second direction when the position of θ**2** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**2**). Subsequently, the rotation mechanism **447** switches the rotation direction of the substrate holder **440** to rotate the substrate holder **440** in the first direction when the position of θ**1** of the substrate Wf reaches the center of the shielding member **481** (when the rotation angle of the specific portion α is θ**1**).

[0064] The shielding mechanism **485** (rotation drive mechanism **486**) switches the rotation direction of the substrate holder **440**, thereby allowing pushing out the shielding member **481** to the shielding position for a period of about five times as illustrated in FIG. **10** compared with a case where the substrate holder **440** is rotated only in the first direction at the constant speed.

[0065] Accordingly, with the embodiment, the electric field in the specific portion α of the substrate Wf can be further strongly suppressed. Additionally, according to the embodiment, since the stirring strength of the plating solution housed in the plating tank **410** can be enhanced by repeatedly rotating the substrate holder **440** in the opposite direction, making the plating film-thickness uniform can be improved.

[0066] While the case where the additional suppression of the deposition rate of plating in the specific portion α of the substrate Wf is performed once or twice (the rotation direction of the substrate holder is switched twice or four times) is described in the above-described embodiment, the number of suppression of the deposition rate of plating in the specific portion α (the number of switching of the rotation direction of the substrate holder) can be appropriately set depending on the degree of suppressing the electric field in the specific portion of the substrate. While an example of the case where one specific portion α is provided to the substrate is described in the above-described embodiment, it is not limited to this, and a plurality of specific portions may be provided. In this case, the shielding mechanism **485** can move the shielding member **481** to the shielding position when the rotation angle is within the predetermined range for each of the plurality of specific portions. The rotation mechanism **447** can switch the rotation direction of the substrate holder **440** between the first direction and the second direction when the rotation angle is within the predetermined range for each of the plurality of specific portions.

[0067] While the example in which the rotation mechanism **447** switches the rotation direction of the substrate holder **440** when the rotation angle of the specific portion of the substrate Wf is within the predetermined range is described in the above-described embodiment, it is not limited to this. For example, the rotation mechanism **447** can switch the rotation direction of the substrate holder **440** when the rotation angle of the specific portion of the substrate Wf is out of the predetermined range. That is, when the electric field in the specific portion of the substrate Wf is to be suppressed, the rotation mechanism **447** can switch the rotation direction of the substrate holder **440** so as to lengthen the period in which the rotation angle of the specific portion of the substrate Wf is within the predetermined range. On the other hand, when the deposition rate of plating in the specific portion of the substrate Wf is to be increased, the rotation mechanism **447** can switch the rotation direction of the substrate holder **440** so as to shorten the period in which the rotation angle of the specific portion of the substrate Wf is within the predetermined range.

[0068] While the example in which the shielding mechanism **485** includes the cam member **487**, the rotation drive mechanism **486**, the driven member **488**, and the like is described in the above-described embodiment, it is not limited to this. The following describes other embodiments of the shielding mechanism **485**.

[0069] FIG. **11** is a perspective view schematically illustrating a configuration of a shielding mechanism of one embodiment. FIG. **12** is a perspective view schematically illustrating the configuration of the shielding mechanism of one embodiment. FIG. **13** is a perspective view schematically illustrating a part of the configuration of the shielding mechanism of one embodiment. FIG. **14** includes plan views schematically illustrating the configuration of the shielding mechanism of one embodiment. FIG. **14**A illustrates a state where the shielding member **481** is at the retracted position, and FIG. **14**B illustrates a state where the shielding member **481** is at the shielding position.

[0070] As illustrated in FIG. **11** to FIG. **14**, the shielding mechanism **485** includes a belt **492** wound around a first pulley **492-1** and a second pulley **492-2**, and a rotation drive mechanism **491** configured to rotate the belt **492** by rotating the first pulley **492-1**. The rotation drive mechanism **491** can be achieved by a known mechanism, such as a rotation motor. The shielding mechanism **485** includes an eccentric cam member **493** as one configuration of a cam member coupled to the second pulley **492-2**. The eccentric cam member **493** is configured to rotate about a rotation shaft **493***a* in association with the rotation of the second pulley **492-2**. The shielding mechanism **485** includes a driven cam member **494** as one configuration of a driven member configured to push out the shielding member **481** to the shielding position in response to pressing by a protrusion **493***b* of the eccentric cam member **493**. Specifically, a bracket **495-1** is attached to the driven cam member **494**, and shafts **495-2** extending in the horizontal direction are attached to the bracket **495-1**. Linear motion guides **496** are attached to the shafts **495-2**. The shielding member **481** is attached to the shafts **495-2** via the plate-shaped bracket **483** extending in the vertical direction.

[0071] With this, as illustrated in FIG. **14**B, when the eccentric cam member **493** rotates to press the driven cam member **494** to a first direction by the protrusion **493***b* of the eccentric cam member **493**, the shielding member **481** is pushed out to the shielding position via the shafts **495-2** and the bracket **483**. When the rotation drive mechanism **491** stops the rotation of the eccentric cam member **493** in this state, the shielding member **481** remains to be moved to the shielding position. On the other hand, the driven cam member **494** is configured to be pushed back to a second direction opposite to the first direction when the driven cam member **494** is not pressed by the protrusion **493***b* of the eccentric cam member **493**. With this, as illustrated in FIG. **14**A, once the eccentric cam member **493** further rotates to release the pressing of the driven cam member **494** by the protrusion **493***b* of the eccentric cam member **493**, the shielding member **481** is pushed back to the retracted position.

[0072] The rotation drive mechanism **491** is configured to rotate the first pulley **492-1** depending on the rotation angle of the substrate holder **440**. That is, similarly to the above-described embodiment, for example, the rotation drive mechanism **491** can rotate the first pulley **492-1** so as to push out the shielding member **481** to the shielding position when the specific portion α of the substrate Wf rotates within the predetermined angle range. This allows for covering the specific portion α of the substrate Wf by the shielding member **481**. Further, the rotation drive mechanism **491** can rotate the first pulley **492-1** so as to cause the shielding member **481** to return to the retracted position when the specific portion α rotates outside the predetermined angle range. With this embodiment, the specific portion α can be covered with the shielding member **481** at a desired timing instead of constantly covering the specific portion α with the shielding member **481**. Additionally, the rotation mechanism **447** can switch the rotation direction of the substrate holder **440** between the first direction and the second direction when the specific portion α of the substrate Wf is within the predetermined range of the rotation angle. Accordingly, since the period of shielding the electric field in the specific portion α can be appropriately controlled, and the stirring strength of the plating solution can be enhanced, the plating film-thickness can be made uniform.

[0073] FIG. **15** is a perspective view schematically illustrating the configuration of the shielding mechanism of one embodiment. FIG. **16** includes plan views schematically illustrating the configuration of the shielding mechanism of one embodiment. FIG. **16**A illustrates a state where the shielding member **481** is at the retracted position, and FIG. **16**B illustrates a state where the shielding member **481** is at the shielding position.

[0074] As illustrated in FIG. **15** and FIG. **16**, the shielding mechanism **485** includes a linear motion drive mechanism **497** configured to linearly move the shielding member **481** between the shielding position and the retracted position. Specifically, the linear motion drive mechanism **497** includes a slider **497***a* configured to reciprocate in the horizontal direction in response to driving of the linear motion drive mechanism **497**. The shielding member **481** is attached to the slider **497***a* via the plate-shaped bracket **483** extending in the vertical direction. The shielding member **481** can be linearly moved between the shielding position and the retracted position by driving the linear motion drive mechanism **497**. The linear motion drive mechanism **497** can be achieved by a known mechanism, such as a linear motion motor.

[0075] The linear motion drive mechanism **497** is configured to linearly move the shielding member **481** between the shielding position and the retracted position depending on the rotation angle of the substrate holder **440**. That is, similarly to the above-described embodiments, for example, the linear motion drive mechanism **497** is configured to push out the shielding member **481** to the shielding position when the specific portion α of the substrate Wf rotates within the predetermined angle range. This allows for covering the specific portion α of the substrate Wf by the shielding member **481**. Further, the linear motion drive mechanism **497** is configured to cause the shielding member **481** to return to the retracted position when the specific portion α rotates outside the predetermined angle range. With this embodiment, the specific portion α can be covered with the shielding member **481** at a desired timing instead of constantly covering the specific portion α with the shielding member **481**. Additionally, the rotation mechanism **447** can switch the rotation direction of the substrate holder **440** between the first direction and the second direction when the specific portion α of the substrate Wf is within the predetermined range of the rotation angle. Accordingly, since the period of shielding the electric field in the specific portion α can be appropriately controlled, and the stirring strength of the plating solution can be enhanced, the plating film-thickness can be made uniform.

[0076] While in the above-described embodiments, the example in which the shielding mechanism

485 is configured to operate in response to the command signal based on information regarding the rotation angle of the substrate holder **440** input from the control module **800** is described, it is not limited to this. FIG. **17** is a vertical cross-sectional view schematically illustrating a configuration of a plating module of one embodiment, and illustrates a state where a shielding member is retracted.

[0077] As illustrated in FIG. **17**, the plating module **400** includes a shielding member **481** for shielding an electric field formed between the anode **430** and the substrate Wf when arranged between the anode **430** and the substrate Wf. The shielding member **481** may be, for example, a shielding plate formed in a plate shape. The shielding member **481** is inserted into the cathode region **422** passing through a side wall of the plating tank **410**, and a flange **484** is attached to an end portion in a side not inserted into the plating tank **410**. In this embodiment, the shielding member **481** is configured to shield a specific portion of the substrate Wf at a desired timing instead of being constantly arranged between the anode **430** and the substrate Wf. The following describe this respect.

[0078] FIG. **18** is a top view schematically illustrating the configuration of the plating module of one embodiment, and illustrates the state where the shielding member is retracted. As illustrated in FIG. **17** and FIG. **18**, the plating module **400** includes the shielding mechanism **460** configured to move the shielding member **481** to the shielding position corresponding to the rotation angle of the substrate holder **440** by the rotation mechanism **447**. The shielding mechanism **460** includes a cam member **461** attached to the substrate holder **440**. The cam member **461** includes a disc cam **462** attached to an upper surface of the seal ring holder **442**. The shielding mechanism **460** includes a driven link **470** configured to push out the shielding member **481** into between the anode **430** and the substrate Wf in response to pressing by a protrusion **462***a* of the cam member **461** (disc cam **462**).

[0079] The driven link **470** includes a follower **473** that is pressed by the protrusion **462***a* of the disc cam **462** to move to a direction moving away from the substrate holder **440**. A base **472** is attached on an outer wall surface at an upper portion of the plating tank **410**, and the follower **473** is supported by the base **472** so as to be able to reciprocate in a radiation direction centering around the shaft **448**. The follower **473** is a rod-shaped member extending in the radiation direction centering around the shaft **448**. The follower **473** has one end portion to which a first roller **471** that rotates about an axis parallel to the rotation axis of the shaft **448** is attached. The follower **473** has the other end portion to which a second roller **475** that is rotatable about an axis perpendicular to both a direction of the rotation axis of the shaft **448** and the radiation direction centering around the shaft **448** is attached.

[0080] The driven link **470** includes a link **474** that rotates in response to a pressing by the follower **473** to push out the shielding member **481** into between the anode **430** and the substrate Wf. The link **474** is a rod-shaped member and rotatably supported by the base **472** about a rotation shaft **476** disposed in the base **472**. The rotation shaft **476** is a rotation shaft parallel to a rotation axis of the second roller **475**. The link **474** is supported by the base **472** so that one side of the link **474** across the rotation shaft **476** can come in contact with the second roller **475**. To an end portion on the other side of the link **474** across the rotation shaft **476**, a third roller **478** that is rotatable about an axis parallel to the rotation axis of the second roller **475** is attached. The link **474** is supported by the base **472** so that the third roller **478** can come in contact with the flange **484** of the shielding member **481**.

[0081] The driven link **470** includes a pressing member **479** that pushes the shielding member **481** back to the retracted position when the shielding member **481** is not pushed out by the link **474**. While the pressing member **479** is, for example, a helical compression spring having one end portion attached to an outer wall of the plating tank **410** and the other end portion attached to the flange **484** of the shielding member **481**, the pressing member **479** is not limited to this.

[0082] Next, an operation of the shielding member **481** by the shielding mechanism **460** will be described. As illustrated in FIG. **17** and FIG. **18**, when the protrusion **462***a* of the disc cam **462** does not press the first roller **471**, the flange **484** is pressed to the direction moving away from the plating tank **410** by a biasing force of the pressing member **479**. This causes the shielding member **481** to move to the retracted position. Further, when the flange **484** is pressed to the direction moving away from the plating tank **410**, the flange **484** presses the third roller **478**, thereby rotating the link **474** counterclockwise. Then, the one side of the link **474** across the rotation shaft **476** presses the second roller **475** toward the center of the shaft **448**. This causes the follower **473** to move toward the center of the shaft **448**.

[0083] FIG. **19** is a vertical cross-sectional view schematically illustrating the configuration of the plating module of one embodiment, and illustrates a state where the shielding member is moved into between an anode and a substrate. FIG. **20** is a top view schematically illustrating the configuration of the plating module of one embodiment, and illustrates the state where the shielding member is moved into between the anode and the substrate. As illustrated in FIG. **19** and FIG. **20**, when the substrate holder **440** rotates and lies within the range of a predetermined rotation angle, the protrusion **462***a* of the disc cam **462** presses the first roller **471**, thereby moving the first roller **471** to a direction moving away from the center of the shaft **448**. In accordance with this, the follower **473** moves to the direction moving away from the center of the shaft **448** and the second roller **475** presses the one side of the link **474** across the rotation shaft **476**. This causes the link **474** to rotate clockwise, and the third roller **478** presses the flange **484** to a direction approaching the plating tank **410** against the biasing force of the pressing member **479**. As a result, the shielding member **481** is pushed out into between the anode **430** and the substrate Wf. When the substrate holder **440** rotates beyond the predetermined range of the rotation angle, the shielding member **481** moves to the retracted position as described using FIG. **17** and FIG. **18**.

[0084] The embodiment includes the shielding mechanism **460** configured to move the shielding member **481** into between the anode **430** and the substrate Wf depending on the rotation angle of the substrate holder **440** instead of constantly arranging the shielding member **481** between the anode **430** and the substrate Wf. Accordingly, the specific portion α of the substrate Wf to be covered with the shielding member **481** can be shielded at the desired timing. Additionally, the rotation mechanism **447** is configured switch the rotation direction of the substrate holder **440** between the first direction and the second direction when the specific portion α of the substrate Wf is within the predetermined range of the rotation angle. Accordingly, since the period of shielding the electric field in the specific portion α can be appropriately controlled, and the stirring strength of the plating solution can be enhanced, the plating film-thickness can be made uniform.

[0085] While the example in which one protrusion **462***a* of the disc cam **462** is disposed is described in this embodiment, it is not limited to this, and for example, when a plurality of specific portions of the substrate Wf are provided along the circumferential direction of the substrate Wf, a plurality of protrusions **462***a* of the disc cam **462** may be disposed corresponding to the arrangement of the specific portions of the substrate Wf. While the example in which one shielding mechanism **460** is disposed is described in this embodiment, it is not limited to this, and a plurality of shielding mechanisms **460** may be disposed along the circumferential direction of the plating tank **410**. This allows covering the specific portion of the substrate Wf with the shielding member **481** when the specific portion of the substrate Wf is within a plurality of different predetermined ranges of the rotation angle.

[0086] FIG. **21** is a vertical cross-sectional view schematically illustrating a configuration of a plating module of one embodiment. The same reference numerals are attached to configurations similar to those in the embodiments indicated by FIG. **3** to FIG. **20**, and the repeated explanation will be omitted.

[0087] As illustrated in FIG. **21**, the plating module **400** includes a film thickness sensor **498** configured to measure the plating film-thickness of the substrate Wf, and a shielding mechanism **499** configured to move the shielding member **481** to the shielding position based on the plating film-thickness of the substrate Wf measured by the film thickness sensor **498**. The shielding mechanism **499** is configured to operate in response to a command signal based on information regarding the plating film-thickness of the substrate Wf input from the control module **800**. The shielding mechanism **499** may have a structure similar to that of any of the shielding mechanisms **485** illustrated in FIG. **5** to FIG. **16**.

[0088] The film thickness sensor **498** is configured to measure the plating film-thickness of the peripheral edge portion of the surface to be plated of the substrate Wf. The film thickness sensor **498** is attached to the ionically resistive element **450** so as to be opposed to the peripheral edge portion of the substrate Wf. The film thickness sensor **498** can measure the plating film-thickness by scanning the peripheral edge portion while the substrate Wf rotates once. However, the film thickness sensor **498** may be configured to measure the plating film-thickness of the whole surface to be plated of the substrate Wf. For example, the film thickness sensor **498** can employ a distance sensor that measures a distance between the film thickness sensor **498** and the substrate Wf (plating film) or a displacement sensor that measures a displacement of the surface to be plated of the substrate Wf. As the film thickness sensor **498**, a sensor for estimating a formation speed of the plating film-thickness may be employed. As the film thickness sensor **498**, for example, an optical sensor, such as a white light confocal sensor, a potential sensor, a magnetic field sensor, or an eddy current sensor are usable.

[0089] The shielding mechanism **499** is configured to linearly move the shielding member **481** between the retracted position and the shielding position so as to make the plating film-thickness of the peripheral edge portion of the substrate Wf uniform. Specifically, the shielding mechanism **499** is configured to move the shielding member **481** to the retracted position when the rotation angle of a region having a thick plating film-thickness is out of the predetermined range in a case where there is the region having the thick plating film-thickness compared with other regions in the distribution of the plating film-thickness of the peripheral edge portion of the substrate Wf. The shielding mechanism **499** is configured to move the shielding member **481** to the shielding position when the rotation angle of the region having the thick plating film-thickness is within the predetermined range. Accordingly, with this embodiment, the region having the thick plating film-thickness of the substrate Wf can be covered with the shielding member **481**.

[0090] Additionally, the rotation mechanism **447** can switch the rotation direction of the substrate holder **440** between the first direction and the second direction when the region having the thick plating film-thickness is within the predetermined range of the rotation angle. That is, when there is a region having a significantly thick plating film-thickness compared with other regions of the surface to be plated, only moving the shielding member **481** to the shielding position while rotating the substrate holder **440** at a predetermined constant speed possibly fails to sufficiently eliminate unevenness in both plating film-thicknesses. In such a case, by switching the rotation direction of the substrate holder **440**, since the period of shielding the electric field in the region having the thick plating film-thickness can be appropriately controlled, and the stirring strength of the plating solution can be enhanced, the plating film-thickness can be made uniform.

[0091] Next, a plating method using the plating module **400** of the embodiment will be described. FIG. **22** is a flowchart of a plating method using the plating module of one embodiment. The following describes a plating method in which the rotation direction of the substrate holder **440** is not switched as illustrated in FIG. **8** as an example.

[0092] In the plating method, the substrate Wf is installed in the substrate holder **440** (Step **102**). Step **102** can be performed by, for example, placing the substrate Wf with the surface to be plated Wf-a facing downward on the seal ring holder **442** with a robot hand (not illustrated) and the like and pressing the back surface of the substrate Wf by the back plate **444**.

[0093] Subsequently, in the plating method, the substrate holder **440** is lowered into the plating tank **410** by the elevating mechanism **443** (lowering step **104**). Subsequently, in the plating method, the substrate holder **440** is rotated in the first direction by the rotation mechanism **447** (first rotating step **106**).

[0094] Subsequently, in the plating method, the plating process is performed on the surface to be plated Wf-a by applying a voltage between the anode **430** arranged in the plating tank **410** and the substrate Wf held by the substrate holder **440** (plating step **108**).

[0095] Subsequently, in the plating method, when a first specific position θ**1** of the substrate Wf reaches the center of the shielding member **481** (Step **110**), the shielding member **481** is moved to the shielding position (shielding step **112**).

[0096] Subsequently, in the plating method, when a second specific position θ**2** of the substrate Wf reaches the center of the shielding member **481** (Step **114**), the shielding member **481** is moved to the retracted position (retracting step **116**).

[0097] Subsequently, in the plating method, whether to end the plating process or not is determined (Step **118**). In the plating method, for example, when it is determined that the plating process is not to be ended because a predetermined time has not elapsed after the start of the plating process (Step **118**, No), the process returns to Step **110** and is continued.

[0098] Meanwhile, in the plating method, for example, when it is determined that the plating process is to be ended because the predetermined time has elapsed after the start of the plating process (Step **118**, Yes), the voltage application between the anode **430** and the substrate Wf is stopped, thereby stopping the plating process (Step **120**). Subsequently, in the plating method, the rotation of the substrate holder **440** by the rotation mechanism **447** is stopped (Step **122**). Subsequently, in the plating method, the substrate holder **440** is raised by the elevating mechanism **443** (Step **124**). Thus, a sequence of the plating process ends.

[0099] Next, another plating method using the plating module **400** of the embodiment will be described. FIG. **23** is a flowchart of a plating method using the plating module of one embodiment. The following describes a plating method in which the rotation direction of the substrate holder **440** is switched several times as illustrated in FIG. **9** and FIG. **10**.

[0100] In the plating method, a first specific position θ**1** and a second specific position θ**2** of the substrate Wf, and a number N of repeating the additional suppression of the deposition rate of plating are set (Step **202**). Values of θ**1**, θ**2**, and the number N of repeating the additional suppression are preferably estimated using an electric field analysis based on the shape of the shielding member **481**, the shape of the specific portion α, and the intensity necessary for suppressing the deposition rate of plating. A plurality of regions in which the deposition rate of plating needs to be suppressed are present in one substrate in some cases. In this case, θ**1**, θ**2**, and the number N of repeating the additional suppression are set for each region.

[0101] Subsequently, the substrate Wf is installed in the substrate holder **440** (Step **204**). Subsequently, in the plating method, the substrate holder **440** is lowered into the plating tank **410** by the elevating mechanism **443** (lowering step **206**). Subsequently, in the plating method, the substrate holder **440** is rotated in the first direction by the rotation mechanism **447** (first rotating step **208**).

[0102] Subsequently, in the plating method, the plating process is performed on the surface to be plated Wf-a by applying a voltage between the anode **430** arranged in the plating tank **410** and the substrate Wf held by the substrate holder **440** (plating step **210**).

[0103] Subsequently, in the plating method, when a first specific position θ**1** of the substrate Wf reaches the center of the shielding member **481** (Step **212**), the shielding member **481** is moved to the shielding position (shielding step **214**).

[0104] Subsequently, in the plating method, when a second specific position θ**2** of the substrate Wf reaches the center of the shielding member **481** (Step **216**), whether N is N=0 or not is determined (Step **218**). In the plating method, when N is determined not to be N=0 (Step **218**, No), the rotation direction of the substrate holder **440** is switched between the first direction and the second direction (inverting step **220**). Specifically, in the inverting step **220**, the rotation speed of the substrate holder **440** is decreased, and the rotation direction of the substrate holder **440** is switched to the

second direction.
[0105] Subsequently, in the plating method, the substrate holder **440** is rotated in the second direction by the rotation mechanism **447** (second rotating step **222**). Subsequently, in the plating method, when the first specific position θ**1** of the substrate Wf reaches the center of the shielding member **481** (Step **224**), the rotation direction of the substrate holder **440** is switched between the first direction and the second direction (inverting step **226**). Specifically, in the inverting step **226**, the rotation speed of the substrate holder **440** is decreased, and the rotation direction of the substrate holder **440** is switched to the first direction.
[0106] Subsequently, in the plating method, the substrate holder **440** is rotated in the first direction by the rotation mechanism **447** (first rotating step **228**). Subsequently, in the plating method, N is decremented (value of N is decreased by 1) (Step **230**). Subsequently, in the plating method, the process returns to Step **216** and is continued. Thus, the additional suppression of the deposition rate of plating in the specific portion α of the substrate Wf is repeated.
[0107] Meanwhile, in the plating method, when N is determined to be N=0 (Step **218**, Yes), the shielding member **481** is moved to the retracted position (retracting step **232**). Subsequently, in the plating method, whether to end the plating process or not is determined (Step **234**). In the plating method, for example, when it is determined that the plating process is not to be ended because a predetermined time has not elapsed after the start of the plating process (Step **234**, No), the process returns to Step **212** and is continued.
[0108] Meanwhile, in the plating method, for example, when it is determined that the plating process is to be ended because the predetermined time has elapsed after the start of the plating process (Step **234**, Yes), the voltage application between the anode **430** and the substrate Wf is stopped, thereby stopping the plating process (Step **236**). Subsequently, in the plating method, the rotation of the substrate holder **440** by the rotation mechanism **447** is stopped (Step **238**). Subsequently, in the plating method, the substrate holder **440** is raised by the elevating mechanism **443** (Step **240**). Thus, a sequence of the plating process ends.
[0109] With the plating method of the embodiment, the specific portion of the substrate Wf can be covered with the shielding member **481** at a desired timing. Additionally, the rotation direction of the substrate holder **440** is switched between the first direction and the second direction when the specific portion of the substrate Wf is within the predetermined range of the rotation angle. Accordingly, since the period of shielding the electric field in the specific portion of the substrate Wf can be appropriately controlled, and the stirring strength of the plating solution can be enhanced, the plating film-thickness of the surface to be plated can be made uniform.
[0110] While some embodiments of the present invention have been described above, the above-described embodiments of the invention are for ease of understanding the present invention, and are not for limiting the present invention. It is obvious that the present invention can be changed or improved without departing from its gist, and that the present invention encompasses its equivalents. Within a range that can solve at least a part of the above-described problems or a range that provides at least a part of the effects, any combination or omission of each component described in the claim and the description are allowed.
[0111] This application discloses, as one embodiment, a plating apparatus that includes: a plating tank for housing a plating solution; an anode disposed in the plating tank; a substrate holder for holding a substrate with a surface to be plated facing downward; a rotation mechanism configured to rotate the substrate holder in a first direction and a second direction opposite to the first direction; and a shielding mechanism configured to move a shielding member into between the anode and the substrate depending on a rotation angle of the substrate holder.
[0112] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the shielding mechanism is configured to move the shielding member into between the anode and a specific portion of the substrate when a rotation angle of the specific portion of the substrate held by the substrate holder is within a predetermined range, and the rotation mechanism is configured

to switch a rotation direction of the substrate holder between the first direction and the second direction when the rotation angle of the specific portion of the substrate is within a predetermined range.

[0113] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the rotation mechanism is configured to switch the rotation direction of the substrate holder between the first direction and the second direction several times when the rotation angle of the specific portion of the substrate is within the predetermined range.

[0114] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the shielding mechanism includes a cam member, a rotation drive mechanism configured to rotate the cam member, and a driven member configured to push out the shielding member to a shielding position between the anode and the substrate in association with the rotation of the cam member.

[0115] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the cam member includes a cam main body configured to be rotated by the rotation drive mechanism, and a rotor attached to the cam main body, and the driven member includes a driven slider provided with a cam groove in which the rotor is fitted, the driven slider is configured to linearly move the shielding member between the shielding position and a retracted position apart from between the anode and the substrate by a pressing from the rotor in association with the rotation of the cam main body.

[0116] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the shielding mechanism further includes a belt wound around a first pulley and a second pulley, the cam member includes an eccentric cam member coupled to the second pulley, the rotation drive mechanism is configured to rotate the eccentric cam member by rotating the first pulley, and the driven member includes a driven cam member configured to push out the shielding member to the shielding position in response to a pressing by a protrusion of the eccentric cam member.

[0117] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the shielding mechanism includes a linear motion drive mechanism configured to linearly move the shielding member between a shielding position and a retracted position, the shielding position is between the anode and the substrate, the retracted position is apart from between the anode and the substrate.

[0118] Furthermore, this application discloses, as one embodiment, the plating apparatus in which the shielding mechanism includes: a cam member attached to the substrate holder, and a driven link configured to push out the shielding member into between the anode and the substrate in response to a pressing by a protrusion of the cam member.

[0119] Furthermore, this application discloses, as one embodiment, a plating method that includes: a lowering step of lowering a substrate holder holding a substrate with a surface to be plated facing downward in a plating tank; a plating step of performing a plating process on the surface to be plated of the substrate lowered in the plating tank; a first rotating step of rotating the substrate holder in a first direction; a second rotating step of rotating the substrate holder in a second direction opposite to the first direction; and a shielding step of moving a shielding member into between an anode and a substrate depending on a rotation angle of the substrate holder.

[0120] Furthermore, this application discloses, as one embodiment, the plating method in which the shielding step is configured to move the shielding member into between the anode and a portion of the substrate when a rotation angle of a specific portion of the substrate held by the substrate holder is within a predetermined range, and the method further includes an inverting step of switching a rotation direction of the substrate holder between the first direction and the second direction when the rotation angle of the specific portion of the substrate is within a predetermined range.

[0121] Furthermore, this application discloses, as one embodiment, the plating method in which the inverting step is configured to switch the rotation direction of the substrate holder between the first direction and the second direction several times when the rotation angle of the specific portion of the substrate is within the predetermined range.

REFERENCE SIGNS LIST

[0122] **400** . . . plating module [0123] **410** . . . plating tank [0124] **430** . . . anode [0125] **440** . . . substrate holder [0126] **443** . . . elevating mechanism [0127] **447** . . . rotation mechanism [0128] **460**, **485**, **499** . . . shielding mechanism [0129] **461** . . . cam member [0130] **470** . . . driven link [0131] **481** . . . shielding member [0132] **486**, **491** . . . rotation drive mechanism [0133] **487** . . . cam member [0134] **488** . . . driven member [0135] **492** . . . belt [0136] **492-1** . . . first pulley [0137] **492-2** . . . second pulley [0138] **493** . . . eccentric cam member [0139] **494** . . . driven cam member [0140] **497** . . . linear motion drive mechanism [0141] **1000** . . . plating apparatus [0142] Wf . . . substrate [0143] Wf-a . . . surface to be plated

Claims

1. A plating method comprising: a lowering step of lowering a substrate holder holding a substrate with a surface to be plated facing downward in a plating tank; a plating step of performing a plating process on the surface to be plated of the substrate lowered in the plating tank; a first rotating step of rotating the substrate holder in a first direction; a second rotating step of rotating the substrate holder in a second direction opposite to the first direction; and a shielding step of moving a shielding member into between an anode and a substrate depending on a rotation angle of the substrate holder.

2. The plating method according to claim 1, wherein the shielding step is configured to move the shielding member into between the anode and a portion of the substrate when a rotation angle of a specific portion of the substrate held by the substrate holder is within a predetermined range, and the method further includes an inverting step of switching a rotation direction of the substrate holder between the first direction and the second direction when the rotation angle of the specific portion of the substrate is within a predetermined range.

3. The plating method according to claim 2, wherein the inverting step is configured to switch the rotation direction of the substrate holder between the first direction and the second direction several times when the rotation angle of the specific portion of the substrate is within the predetermined range.